

$$10,000 < N / C < 1,000,000$$

N = total number of unknowns = problem size = $m \cdot n$

C = total number of cores = nodes ppn * threads

N / C = unknowns per core

ppn = processes per node = MPI ranks per node

threads = threads per MPI rank = [1, 2, 4, 8]

nodes = [1, 2, 4]

maxCores = 128

nodes	ppn	MPI ranks	T=1 N range	T=2 N range	T=4 N range	T=8 N range
1	1	1	10k-1m	20k-2m	40k-4m	80k-8m
1	2	2	20k-2m	40k-4m	80k-8m	160k-16m
1	4	4	40k-4m	80k-8m	160k-16m	320k-32m
1	8	8	80k-8m	160k-16m	320k-32m	640k-64m
1	16	16	160k-16m	320k-32m	640k-64m	1.28m-128m
1	32	32	320k-32m	640k-64m	1.28m-128m	invalid
1	64	64	640k-64m	1.28m-128m	invalid	invalid
1	128	128	1.28m-128m	invalid	invalid	invalid
2	1	2	20k-2m	40k-4m	80k-8m	160k-16m
2	2	4	40k-4m	80k-8m	160k-16m	320k-32m
2	4	8	80k-8m	160k-16m	320k-32m	640k-64m
2	8	16	160k-16m	320k-32m	640k-64m	1.28m-128m
2	16	32	320k-32m	640k-64m	1.28m-128m	invalid
2	32	64	640k-64m	1.28m-128m	invalid	invalid
2	64	128	1.28m-128m	invalid	invalid	invalid
4	1	4	40k-4m	80k-8m	160k-16m	320k-32m
4	2	8	80k-8m	160k-16m	320k-32m	640k-64m
4	4	16	160k-16m	320k-32m	640k-64m	1.28m-128m
4	8	32	320k-32m	640k-64m	1.28m-128m	invalid
4	16	64	640k-64m	1.28m-128m	invalid	invalid
4	32	128	1.28m-128m	invalid	invalid	invalid

Approx N	problemSize 2D	Actual 2D N	problemSize 3D	Actual 3D N
10k	100 × 100	10,000	22 × 22 × 22	10,648
100k	316 × 316	99,856	46 × 46 × 46	97,336
250k	512 × 512	262,144	64 × 64 × 64	262,144
1m	1024 × 1024	1,048,576	128 × 128 × 128	2,097,152
2m	1414 × 1414	1,999,396	126 × 126 × 126	2,000,376
4m	2048 × 2048	4,194,304	160 × 160 × 160	4,096,000
10m	3162 × 3162	9,998,244	215 × 215 × 215	9,938,375
16m	4096 × 4096	16,777,216	256 × 256 × 256	16,777,216
64m	8192 × 8192	67,108,864	400 × 400 × 400	64,000,000
128m	11314 × 11314	127,999,396	504 × 504 × 504	128,024,064